

12.14.

$$r_{T_2 \text{ med } T_1} = \ln \left(\frac{T_2}{T_1} \right) \sim 1K$$

麦克斯韦速率分布函数:

$$f(v) = 4\pi \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}}$$

$$v_p = \sqrt{\frac{2kT}{m}}$$

$$f(v_p) = \frac{4}{\sqrt{\pi}} \left(\frac{m}{2kT} \right)^{\frac{3}{2}} v_p^2 \cdot e^{-1} = \frac{4}{\sqrt{\pi} v_p e}$$

$$\frac{\Delta N}{N} \approx f(v_p) \cdot \Delta v = \frac{0.04}{\sqrt{\pi} e} \approx 0.83\%$$

12.17.

$$(1) \frac{v_1}{v_2} = \sqrt{\frac{m_{O_2}}{m_H}}$$

$$(2) E = \frac{3}{2} PV = \frac{3}{2} PV$$

$$\text{可得 } P_1 = \frac{2E_0}{3V}, \quad P_2 = \frac{2E_0}{3V}$$

$$P = P_1 + P_2 = \frac{4E_0}{3V}$$

2.18

$$Nf(v) = \begin{cases} \frac{a}{V_0} v, & 0 \leq v \leq V_0 \\ a, & V_0 < v \leq 2V_0 \\ 0, & v > 2V_0 \end{cases}$$

$$(1) \int_0^{2V_0} Nf(v) dv = \frac{1}{2} V_0 \cdot a + (2V_0 - V_0) \cdot a = \frac{3}{2} V_0 a = N$$

$$a = \frac{2N}{3V_0}$$

$$\bar{\epsilon} = \frac{1}{N} \int_0^{2V_0} \frac{1}{2} m_0 v^2 \cdot Nf(v) dv = \frac{31}{36} m_0 V_0^2$$

$$(2) \Delta N = \int_{\frac{V_0}{2}}^{\frac{3V_0}{2}} Nf(v) dv = \int_{\frac{V_0}{2}}^{V_0} \frac{a}{V_0} v dv + \int_{V_0}^{\frac{3V_0}{2}} a dv = \frac{3}{8} V_0 a + \frac{1}{2} V_0 a = \frac{7}{8} V_0 a = \frac{7}{12} N$$

12.21

$$\lambda = \frac{kT}{\sqrt{2\pi} d^2 P}$$

$$P = \frac{kT}{\sqrt{2\pi} d^2 \lambda} < \frac{1.38 \times 10^{-23} \times 300}{\sqrt{2} \times 3.14 \times (3.5 \times 10^{-10})^2 \times 0.07} \approx 10^8 \text{ Pa}$$

应降低压强到约 0.109 Pa

12.25.

$$K = \frac{1}{3} \bar{v} \cdot C_v = \frac{2k\bar{v}}{6\sqrt{2\pi} d^2}$$

$$\bar{v} = \sqrt{\frac{8 \times 8.314 \times 273.15}{3.1416 \times 0.028}} \approx 454.4 \text{ m/s}$$

$$d = \sqrt{\frac{2k\bar{v}}{6\sqrt{2\pi} K}} \approx 2.23 \times 10^{-10} \text{ m}$$

13.4

$$\Delta E = \frac{1}{2} (P_B V_B - P_A V_A)$$

$$= \frac{5}{2} (6 \times 10^5 \times 4.5 \times 10^{-3} - 8.2 \times 10^5 \times 3 \times 10^{-3}) = 600 \text{ J}$$

$$(1) W_{ACB} = W_{AC} + W_{CB} = W_{CB}$$

$$= P_2 (V_2 - V_1) = 6 \times 10^5 \times (4.5 - 3) \times 10^{-3} = 900 \text{ J}$$

$$Q_{ACB} = \Delta E + W = 1500 \text{ J}$$

$$(2) W_{ADB} = W_{AD} + W_{DB} = W_{AD}$$

$$= P_1 (V_2 - V_1) = 1230 \text{ J}$$

$$\bar{\epsilon}' = \frac{1}{\Delta N} \int_{\frac{V_0}{2}}^{\frac{3V_0}{2}} \frac{1}{2} m_0 v^2 \cdot Nf(v) dv = \frac{197}{336} m_0 V_0^2$$



13.6

$$(1) W=0, Q=\Delta U+W=\Delta U$$

$$\Delta T = \frac{Q}{n \cdot \frac{5}{2} R} = \frac{500}{2 \cdot \frac{5}{2} \cdot 8.314} \approx 12.03 \text{ K}$$

$$T_1 = T_0 + \Delta T = 273.15 + 12.03 = 285.18 \text{ K}$$

$$(2) \Delta U=0, Q=W$$

$$W = nRT_0 \ln\left(\frac{V_2}{V_0}\right) = Q$$

$$500 = 2 \times 8.314 \times 273.15 \times \ln\left(\frac{V_2}{V_0}\right)$$

$$\Rightarrow \frac{V_2}{V_0} = e^{0.110} \approx 1.116$$

$$V_2 = 1.116 \cdot 44.8 \text{ L} \approx 50.0 \text{ L}$$

$$P_2 = P_0 \cdot \frac{V_0}{V_2} = \frac{1.013 \times 10^5}{1.116} \approx 0.908 \times 10^5 \text{ Pa}$$

$$(3) Q=\Delta U+W$$

$$\Delta U = Q \cdot \frac{C_V}{C_P} = \frac{5}{7} Q \approx 357.1 \text{ J}$$

$$W = \frac{2}{7} Q \approx 142.9 \text{ J}$$

$$\Delta T = \frac{Q}{n \cdot \frac{5}{2} R} \approx 8.59 \text{ K}$$

$$T_3 = T_0 + \Delta T = 281.74 \text{ K}$$

$$V_3 = V_0 \cdot \frac{T_3}{T_0} \approx 46.2 \text{ L}$$

13.11

$$TV^{\gamma-1} = \text{常数}$$

$$\bar{V} = \sqrt{\frac{8RT}{\pi M}} \Rightarrow \bar{V} \propto \sqrt{T}$$

$$\frac{\bar{V}_2}{\bar{V}_1} = \sqrt{\frac{T_2}{T_1}}$$

$$\frac{T_2}{T_1} = \left(\frac{\bar{V}_2}{\bar{V}_1}\right)^2 = 2^2 = 4$$

$$\frac{\bar{V}_2}{\bar{V}_1} = \sqrt{2} = 2^{\frac{\gamma-1}{2}}$$

$$(1) \text{ 对于单原子分子, } \gamma = \frac{5}{3}, \gamma = \frac{5}{3}$$

$$\frac{\bar{V}_2}{\bar{V}_1} = \sqrt{2} \approx 1.26$$

$$(2) \text{ 对于双原子分子, } \gamma = \frac{7}{5}, \gamma = \frac{7}{5}$$

$$\frac{\bar{V}_2}{\bar{V}_1} = \sqrt[5]{2} \approx 1.15$$

$$3.8 (1) \text{ 等容升温: } W_{AB}=0$$

$$Q_{AB} = \Delta U_{AB}$$

$$= \frac{5}{2} nR(T_2 - T_1)$$

$$= \frac{5}{2} \times 1 \times 8.314 \times (355.15 - 273.15)$$

$$= 1247.1 \text{ J}$$

$$\text{等压膨胀: } \Delta U_{BC}=0$$

$$W_{BC} = nRT_2 \ln\left(\frac{2V_0}{V_0}\right) \approx 2035.2 \text{ J}$$

$$Q_{BC} = W_{BC} = 2035.2 \text{ J}$$

$$W_1 = 2035.2 \text{ J}$$

$$Q_1 = 1247.1 + 2035.2 = 3282.3 \text{ J}$$

$$\Delta U_1 = 1247.1 \text{ J}$$

$$(2) \text{ 等压膨胀:}$$

$$\Delta U_{AD}=0$$

$$W_{AD} = nRT_1 \ln\left(\frac{2V_0}{V_0}\right) \approx 1689.4 \text{ J}$$

$$Q_{AD} = W_{AD} = 1689.4 \text{ J}$$

$$\text{等容升温}$$

$$W_{DC}=0$$

$$Q_{DC} = \Delta U_{DC} = 1247.1 \text{ J}$$

$$W_2 = 1689.4 \text{ J}$$

$$Q_2 = 1689.4 + 1247.1 = 2936.5 \text{ J}$$

$$\Delta U_2 = 1247.1 \text{ J}$$



13.16.

a → b: $Q_{ab} = Q_{ba} = nC_V(T_b - T_a)$

$T_b = \frac{P_1 V_1}{nR}, T_a = \frac{P_2 V_1}{nR}$

$Q_{ab} = \frac{C_V}{R} V_1 (P_1 - P_2) = \frac{V_1 (P_1 - P_2)}{\gamma - 1}$

b → c: $Q = 0$

c → a: $Q_{ca} = -Q_{ac} = nC_p(T_c - T_a)$
 $= \frac{C_p}{R} P_2 (V_2 - V_1)$
 $= \frac{\gamma P_2 (V_2 - V_1)}{\gamma - 1}$

$\eta = \frac{Q_{ab} - Q_{ca}}{Q_{ab}} = 1 - \frac{Q_{ca}}{Q_{ab}}$
 $= 1 - \frac{\gamma (V_2 - V_1) / V_1}{(P_1 - P_2) / P_2}$
 $= 1 - \frac{\gamma (V_2 / V_1) - 1}{(P_1 / P_2) - 1}$

3.18.

B → C 绝热压缩, $Q_{BC} = 0$

$T_C = T_B \left(\frac{V_1}{V_2} \right)^{\gamma-1}$

C → D 等压膨胀, $P_C = P_D$

$T_D = T_C \frac{V_1}{V_2}$

$Q_{CD} = nC_p(T_D - T_C)$

D → E 绝热膨胀, $Q_{DE} = 0$

$T_E = T_D \left(\frac{V_1}{V_1} \right)^{\gamma-1}$

E → B 等压压缩, $V_E = V_B = V_1$

$Q_{EB} = nC_v(T_E - T_B)$

$\eta = 1 - \frac{nC_v(T_E - T_B)}{nC_p(T_D - T_C)} = 1 - \frac{1}{\gamma} \frac{T_E - T_B}{T_D - T_C}$

$T_E = \left[T_B \left(\frac{V_1}{V_2} \right)^{\gamma-1} \left(\frac{V_1}{V_2} \right) \right] \left(\frac{V_1}{V_1} \right)^{\gamma-1} = T_B \left(\frac{V_1}{V_2} \right)^{\gamma}$

代入得: $\eta = 1 - \frac{(V_1/V_2)^{\gamma} - 1}{\gamma (V_1/V_2)^{\gamma-1} - 1}$

13.25.

11) $\Delta S_{\text{总}} = \int \frac{dQ}{T} = \int_{T_1}^{T_2} \frac{mc dT}{T} = mc \ln \left(\frac{T_2}{T_1} \right)$
 $\approx -163.87 \text{ J/K}$

12) $Q = mc(T_1 - T_2)$
 $= 0.18 \times 4.18 \times 10^3 \times (373.15 - 300.15)$
 $= 54925.2 \text{ J}$

$\Delta S_{\text{总}} = \frac{Q}{T_0} = \frac{54925.2}{300.15} \approx 183.00 \text{ J/K}$

13) $\Delta S_{\text{总}} = \Delta S_{\text{水}} + \Delta S_{\text{空气}}$
 $= 19.13 \text{ J/K}$

